

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination –Summer 2023

Course: S.Y. B. Tech.

Semester: III

Branch: Electrical Engineering/Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. /Electrical & Power Engineering

Subject Code & Name: BTEEC302 Electrical Machines-I

Max Marks: 60

Date: 10/08/2023

Duration: 3Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|--|------------|----------|
| A) Draw and explain equivalent circuit of Transformer | Remember | 6 |
| B) Draw and explain phasor diagram of transformer on Inductive load | Understand | 6 |
| C) Write down formulas and State clearly the meaning of the symbols used and units [2M Each]
1. EMF equation of the transformer
2. Hysteresis loss
3. Eddy current loss | Remember | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|--|------------|----------|
| A) Explain parallel operation of transformer with its needs, advantages, disadvantages and conditions for parallel operation of transformer | Remember | 6 |
| B) Enlist different transformer connection & explain any two. | Understand | 6 |
| C) A 4 KVA 200/400 V, 1Ø transformer has equivalent resistance and reactance referred to low voltage side equal to 0.5 Ω & 1.5 Ω respectively, find the terminal voltage on the high voltage side when it supplies $\frac{3}{4}$ th full load at power factor of 0.8, the supply voltage being 220 V. Hence find the output of the transformer and it's efficiency if the core losses are 100 W | Analyze | 6 |

Q. 3 Solve Any Two of the following.

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|--|------------|----------|
| A) Derive the equation for mechanical force acting on current carrying conductor placed in magnetic field. | Remember | 6 |
| B) Write the equation for force that governs doubly excited magnetic field. | Understand | 6 |
| C) Define the following terms
1. Pole pitch
2. Pitch factor
3. Chording angle | Understand | 6 |

Q.4 Solve Any Two of the following.

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|--|------------|----------|
| A) Derive the EMF equation of the DC Generator.
(State clearly the meaning and units of the symbols used) | Understand | 6 |
|--|------------|----------|

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|-----------|---|----------|----------|
| B) | Explain armature reaction with cross magnetizing and demagnetizing. | Remember | 6 |
| C) | An 8 pole armature has 96 slots with 8 conductors per slot. It is driven at 600 rpm. The useful flux per pole is 10 mWb. Calculate the induced e.m.f. in armature winding when it is: (a) lap connected (b) wave connected. | Analyze | 6 |

Q.5 Solve Any Two of the following.

- | | | | |
|-----------|---|------------|----------|
| A) | Explain the function of No volt coil and overload coil of the 3 point starter | Understand | 6 |
| B) | Explain the permanent magnet stepper motor and their applications. | Remember | 6 |
| C) | Explain construction and working of reluctance motor | Remember | 6 |